MA416

Lecture 08 – September 25, 2025

Overview of One-Factor ANOVA Testing

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Tests on Multiple Variances: $H_0: \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_g^2$

Bartlett Test for Homoscedasticity

Assumptions...

1. Populations are normally distributed
2. Independence between samples
3. Independence of observations within samples

We define:

$$\mathrm{Ba}_{stat} = \frac{\nu \cdot \ln\left(s_p^2\right) - \left[\nu_1 \cdot \ln(s_1^2) + \nu_2 \cdot \ln(s_2^2) + \cdots + \nu_g \cdot \ln\left(s_g^2\right)\right]}{1 + \frac{1}{3 \cdot (g-1)} \cdot \left[\left(\frac{1}{\nu_1} + \frac{1}{\nu_2} + \cdots + \frac{1}{\nu_g}\right) - \frac{1}{\nu}\right]}$$

Where $\nu = n - g, \nu_j = n_j - 1$, and s_p^2 the pooled sample variance.

If all assumptions are met, then:

$$\mathrm{Ba}_{stat} \sim \chi^2_{g-1}$$

Where if H_0 is false, then the ideal value for $\mathrm{Ba}_{stat} \gg 0$.

Tests on Multiple Means: $H_0: \mu_1 = \mu_2 = \cdots = \mu_g$

ANOVA Test for Homogeneity

Assumptions...

1. Populations are normally distributed
2. Populations are homoscedastic to one another
3. Independence between samples
4. Independence of observations within samples

We define:

$$\text{SST} = s_{all}^2 \cdot (n-1) \qquad \text{SSE} = \sum_{j=1}^{g} \nu_j \cdot s_j^2 \qquad \text{SSM} = \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{\bar{x}}\right)^2$$

Where $n = n_1 + n_2 + \cdots + n_g$, $\bar{\bar{x}}$ the grand mean, and s_{all}^2 the grand variance.

In addition, we define:

$$MST = \frac{SST}{n-1} \qquad MSE = \frac{SSE}{n-g} \qquad MSM = \frac{SSM}{g-1} \qquad F_{stat} = \frac{MSM}{MSE}$$

Tests on Multiple Means: $H_0: \mu_1 = \mu_2 = \cdots = \mu_g$

$$MST = \frac{SST}{n-1} \qquad MSE = \frac{SSE}{n-g} \qquad MSM = \frac{SSM}{g-1} \qquad F_{stat} = \frac{MSM}{MSE}$$

If all assumptions are met, then:

$$F_{stat} \sim F_{(g-1,n-g)}$$

Where if H_0 is false, then the ideal value of:

- $0 \ll SSM \approx SST$
- $SSE \approx 0^+$
- $F_{stat} \approx \frac{SST}{0^+} \gg 0$

Tests on Multiple Means: $H_0: \mu_1 = \mu_2 = \cdots = \mu_g$

Example: Suppose you want to see if there is a significant difference in job efficiency across different job ranks at a company. Assume also that the normality assumption is met (if not assumed, you must conduct $g \times$ Lilliefors' tests!!!)

```
mydata = read.csv('CompanyPayroll_9.25.csv')

X = mydata$Efficiency_Factor_Obs
Xs = split(X, mydata$Job_Rank)

# two common R-orderings of a variable; be aware of which
# functions use which orderings (both generated from below)
labels(Xs)
unique(mydata$Job_Rank)

xbarj = sapply(Xs, mean)
s2j = sapply(Xs, var)
nj=sapply(Xs, length)
vj = nj - 1

n = sum(nj)
v = sum(vj)
```

```
g = n - v
sp2 = sum(vj*s2j)/v
Ba_top = v*log(sp2) - sum(vj*log(s2j))
Ba_bot = 1 + 1/(3*(g-1))*(sum(1/vj) - 1/v)

Ba_stat = Ba_top/Ba_bot # approx 3.62
pchisq(Ba_stat, g-1, lower.tail=F) #approx 0.16
```

Tests on Multiple Means: $H_0: \mu_1 = \mu_2 = \cdots = \mu_g$

```
s2all = var(X)
xbarbar = mean(X)

SST = s2all*(n-1)
SSE = sum(s2j*vj)
SSM = sum(nj*(xbarj - xbarbar)^2)
SST - (SSE + SSM) # if not 0, fix it

MST = SST/(n-1)
MSE = SSE/(n-g)
MSM = SSM/(g-1)
Fstat = MSM/MSE
pf(Fstat, g-1, n-g, lower.tail=F) #approx 0.000022
```

Since $p_{val} \approx 0$, we have evidence to believe that at least two means are significantly different from one another (provided that the assumptions are all met). Investigate which with pairwise pooled-variance T-tests for inference as to which are the different ones.

```
> xbarj
   Associate Ast. Manager         Entry
     2.560000     5.133333     2.815217
```